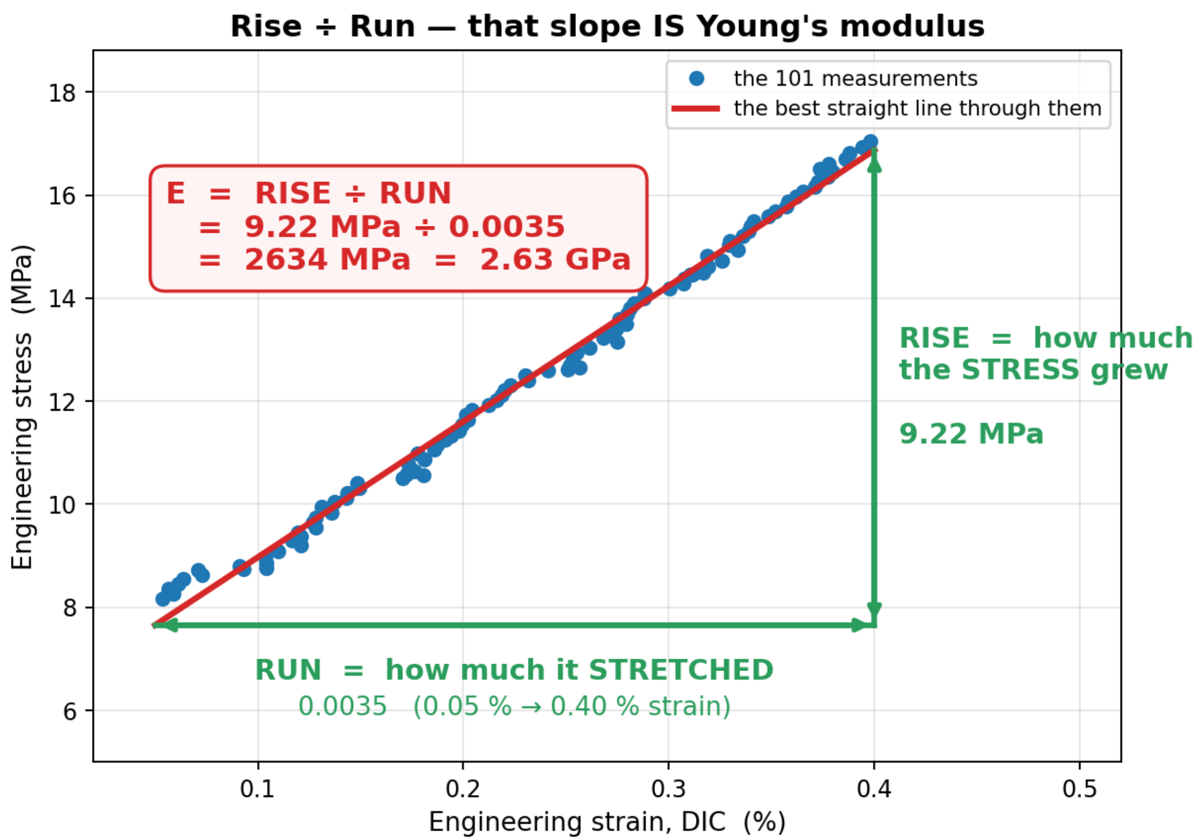
Young's modulus E — the slope of the ELASTIC region

Elastic = the specimen would spring back if you let go. E is the slope of the graph there, and nowhere else.



STRESS = how hard you are pulling ÷ how thick the specimen is. Units MPa.

STRAIN = how much the gauge stretched ÷ how long it was. Just a fraction — no units.

E = the SLOPE of the elastic region: how much the STRESS went up, divided by how much it STRETCHED to get there.

```
RISE  =   16.87   -   7.65   =  9.22 MPa
RUN   =  0.0040  - 0.0005  =  0.0035

E  =  9.22 / 0.0035  =  2634 MPa
```

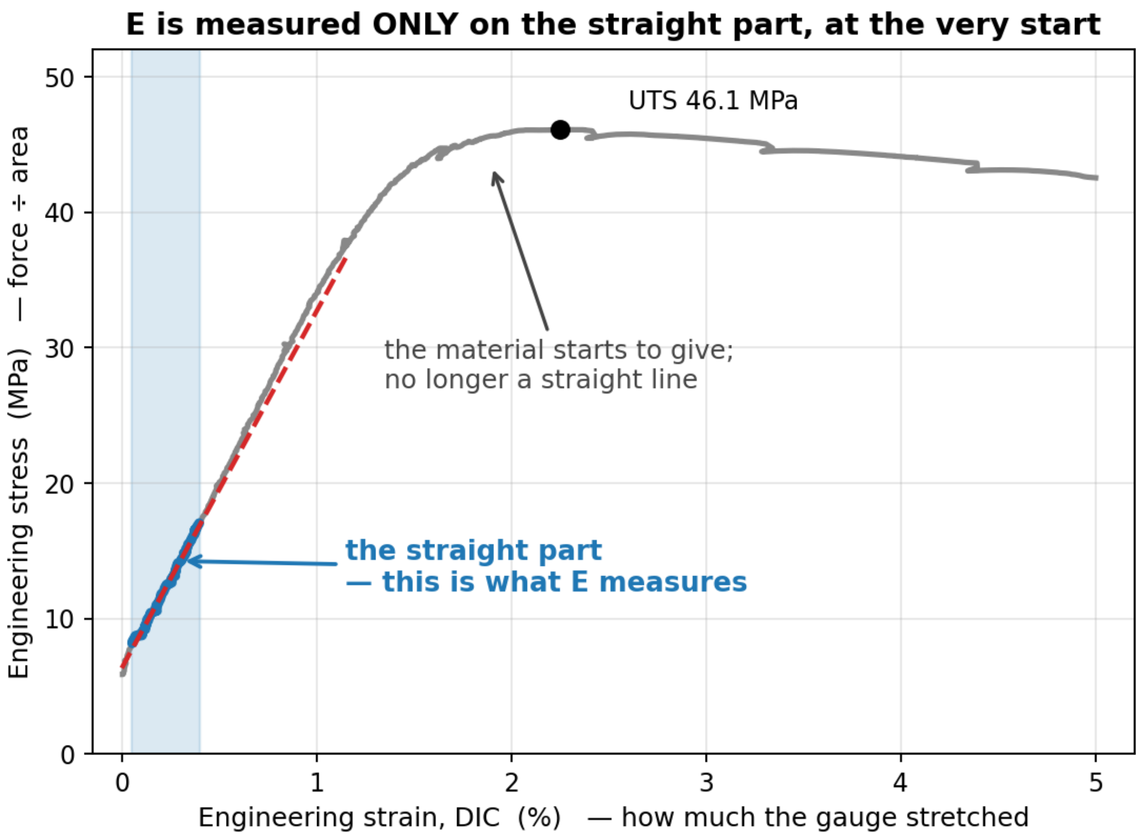
2634 MPa = 2.63 GPa. A stiffer material gives a steeper line and a bigger E; a floppier one gives a shallower line.

Slope of the ELASTIC region = stiffness. Nothing past the elastic region is used for E.

V6d · S11 · 100 % infill — the real measurements, not an illustration

The formula the code actually uses

Two points would do — but 101 were measured, so all 101 are used. That is all "least squares" means.



Best-fit slope through all the points:

$$E = \frac{n\cdot\Sigma(\varepsilon\cdot\sigma) - \Sigma\varepsilon \cdot \Sigma\sigma}{n\cdot\Sigma(\varepsilon\cdot\varepsilon) - (\Sigma\varepsilon)^2}$$

Add up four columns over the 101 points:

```
n          = 101
Σε         = 0.237696
Σσ         = 1266.1435  MPa
Σ(ε·ε)     = 0.000657593
Σ(ε·σ)     = 3.238406  MPa
```

Put the numbers in:

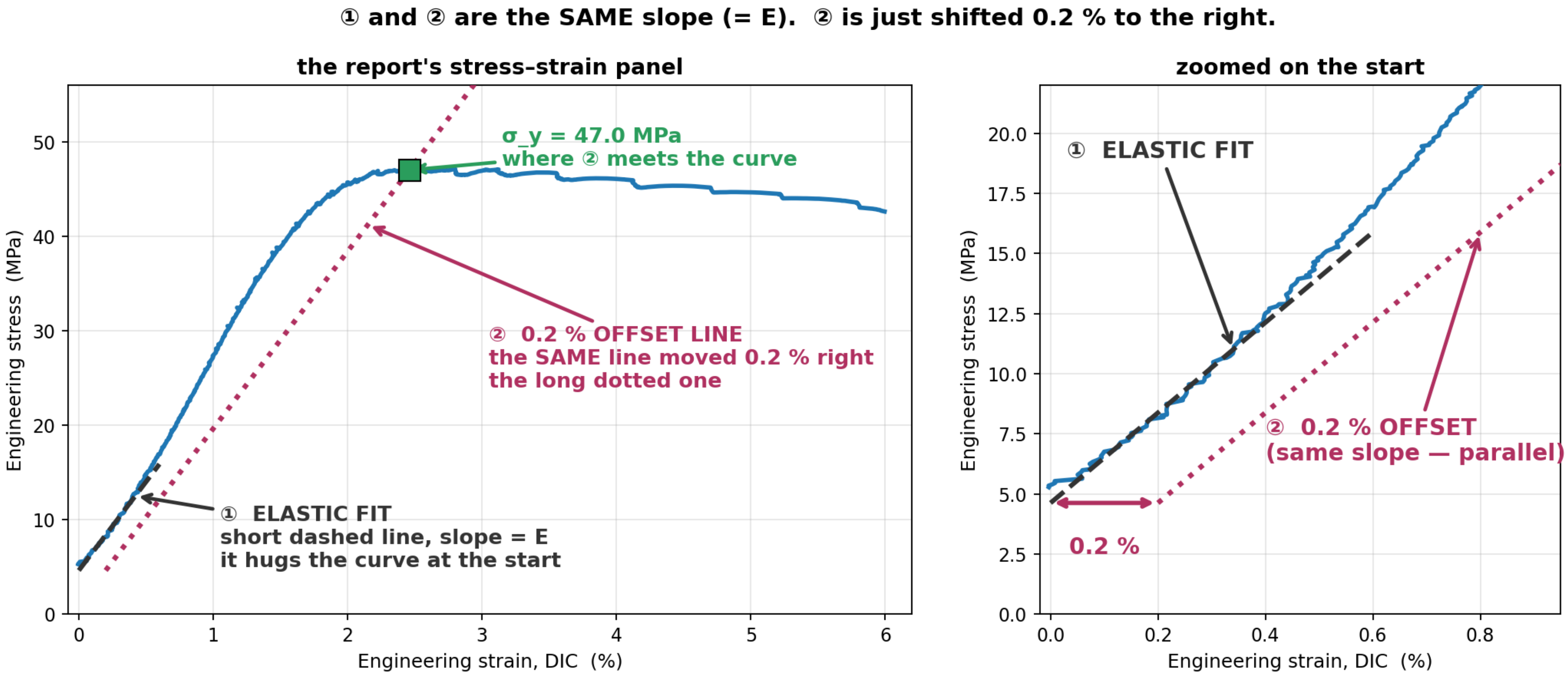
```
top    = 101·3.2384 − 0.2377·1266.14 = 26.1218
bottom = 101·0.0006576 − 0.2377²   = 0.0099175
E = 26.1218 / 0.0099175 = 2634 MPa = 2.63 GPa
```

$\sigma = 2634 \times \varepsilon + 6.34$ $R^2 = 0.9936$ the 6.34 MPa offset is the preload the curve starts from, not an error

utm_analysis.analyze() · fit window 0.05 %–0.40 % strain · σ = (F + 470 N anchor) / 80 mm²

The two grey lines on the report's stress–strain graph

Both have the SAME slope — the one E you measured. Neither of them is used to compute E.



1 Fit E first — least squares over the points from 0.05 % to 0.40 % strain.

2 Draw ① with that slope: the short dashed line that hugs the start of the curve.

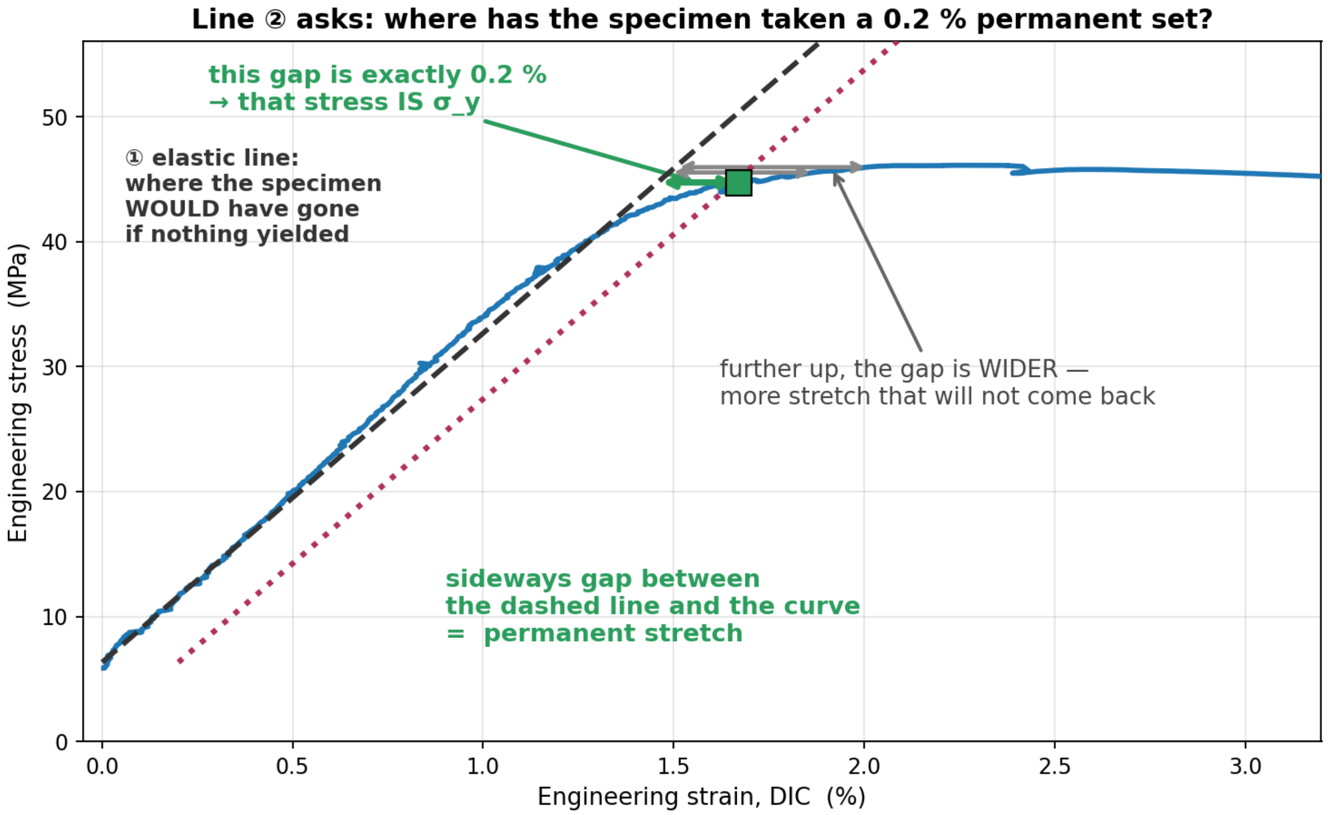
3 Copy it and slide it 0.2 % to the right — that copy is ②, the long dotted line.

4 Where ② meets the curve is σ_y, the 0.2 % offset yield stress.

The line is not an extrapolation FROM the yield point — the yield point is defined BY the line. S16 · E 1.88 GPa · σ_y 47.0 MPa

Why line (2) is needed: PLA has no obvious yield point

Steel snaps to a clear yield. PLA just bends over gradually — so "where did it yield" has to be DEFINED, not read off.



What (2) depicts: the elastic response of a specimen that has ALREADY been permanently stretched by 0.2 %.

Why sideways: at any stress, the horizontal distance from the elastic line to the curve is the stretch that will NOT spring back when you unload.

So σ_y is: the stress at which that permanent stretch first reaches 0.2 %. That is where (2) meets the curve.

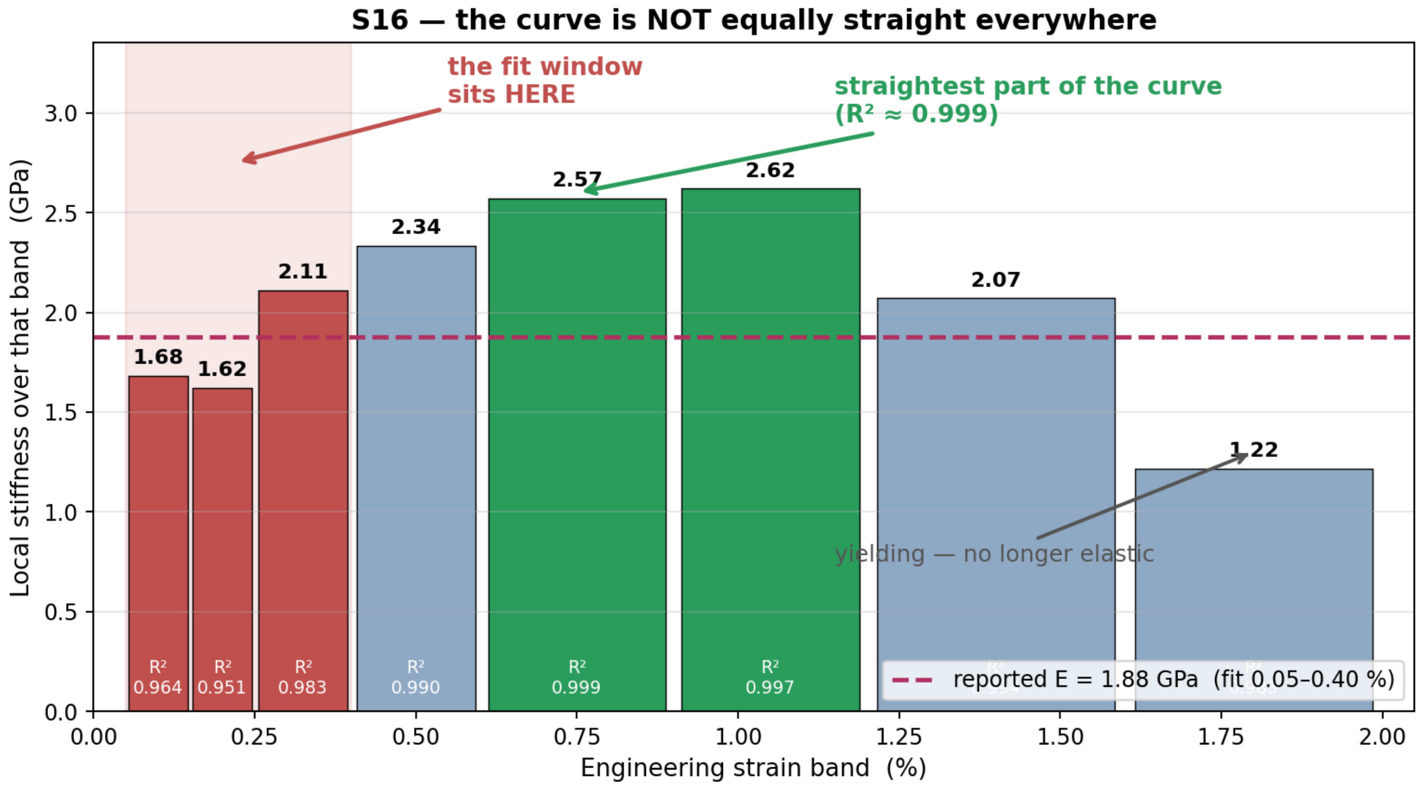
```
elastic line reaches 44.7 MPa at  1.46 %
the CURVE reaches    44.7 MPa at  1.67 %
                 difference   0.21 %  ✓
```

Without this convention there is no yield number at all — the curve never gives you one.

σ_y is a DEFINITION, not a feature of the graph. 0.2 % is the agreed convention (ISO 527, ASTM E8).

Why the fit stops at 0.40 % — and what that costs us

Past the elastic region the curve is no longer straight, so a slope measured there would not be a modulus at all.



Why not further? beyond ~1.2 % the material is yielding. Its slope collapses (2.62 → 2.07 → 1.22 GPa). Including that would average elastic and plastic behaviour together.

Why not from 0 %? the first fraction of a percent is grip take-up and specimen settling, not material response — the noisiest part of the whole test.

0.05–0.40 % is the convention (ISO 527 uses 0.05–0.25 %). It is chosen to sit safely inside the elastic region.

BUT on S16 that window lands on the toe, not the elastic region: it reads 1.62–2.11 GPa, where the straightest part of the curve (0.6–1.2 %, $R^2 \geq 0.997$) reads 2.57–2.62 GPa. Reported E = 1.88 GPa is therefore an UNDER-estimate.

Open question for the campaign: move the window to 0.25–0.60 %, or keep ISO comparability? It changes every published E.

S16 · local slope fitted in each band separately · R^2 printed inside each bar